ClipSAT : Math for AI & Data Science

Unit 4.01: Conditional Probability & Bayes' Rule

Name: _____

Q1. In a dataset, the prior probability of class C_1 is PC_1 = 0.10 . What is the probability of the complement class Pneg C_1 ?

[ ] A: 0.90
[ ] B: 0.95
[ ] C: 0.05
[ ] D: 1.10

Q2. In a dataset, the prior probability of class C_1 is PC_1 = 0.20 . What is the probability of the complement class Pneg C_1 ?

[ ] A: 0.80
[ ] B: 0.90
[ ] C: 0.10
[ ] D: 1.20

Q3. In a dataset, the prior probability of class C_1 is PC_1 = 0.60 . What is the probability of the complement class Pneg C_1 ?

[ ] A: 0.40
[ ] B: 0.70
[ ] C: 0.30
[ ] D: 1.60

Q4. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.05 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[ ] A: 0.0450
[ ] B: 0.9500
[ ] C: 0.8500
[ ] D: 0.0050

Q5. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.10 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[ ] A: 0.0900
[ ] B: 1.0000
[ ] C: 0.8000
[ ] D: 0.0100

Q6. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.15 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[ ] A: 0.1350
[ ] B: 1.0500
[ ] C: 0.7500
[ ] D: 0.0150

ClipSAT : Math for AI & Data Science

Unit 4.01: Conditional Probability & Bayes' Rule

Name: _____

Q7. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.20 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[ ] A: 0.1800
[ ] B: 1.1000
[ ] C: 0.7000
[ ] D: 0.0200

Q8. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.25 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[ ] A: 0.2250
[ ] B: 1.1500
[ ] C: 0.6500
[ ] D: 0.0250